
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

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Abstract

Generative image models trained on large-scale image-text corpora now produce highly realistic images, yet they continue to omit or distort concepts that should plausibly be available in their training data, including human hands and objects appearing in groups of four. Because such failures have primarily been reported through anecdotes or narrowly specified probes, it remains unclear whether they are isolated curiosities or manifestations of a broader distributional limitation. We address this gap by formalizing and measuring *conceptual blindspots*: concepts present in natural data but systematically absent, suppressed, or misrepresented in generated images. Our method uses sparse autoencoders (SAEs) to obtain interpretable concept embeddings, making it possible to compare concept prevalence quantitatively between real and synthesized images. We train an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts—the largest such SAE to date—which supports fine-grained analysis of conceptual disparities at distribution and datapoint levels. Applied to four widely used generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), the framework identifies suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) as well as exaggerated blindspots (e.g., wood background texture and palm trees). At the individual datapoint level, the same representation isolates memorization artifacts—cases in which models reproduce highly specific visual templates seen during training. Together, these results establish a theoretically grounded and empirically scalable framework for assessing the conceptual fidelity of generative image models relative to the underlying data-generating process.

1 Introduction

Generative image models trained on large-scale datasets have reached a level of visual competence that supports applications within computer vision [1–9] and in adjacent domains [9–15]. This progress, however, coexists with a persistent and empirically documented limitation: qualitative [16–18] and quantitative studies [19–20] show that models can fail on concepts that appear simple from a human perspective, including human hands [21–24], objects appearing in groups of four [25], and negations or object relations [20, 26]. When prompted with these concepts, models often generate images containing nearby visual structure rather than the intended concept itself—for example, a hand-like object with six fingers. We refer to these failures as “conceptual blindspots”¹—concepts that are expected to be represented in the training distribution but are absent, suppressed, or distorted in the model’s outputs. The central question is therefore not whether individual prompts can expose surprising errors, but whether such errors instantiate a systematic phenomenon in which

¹We borrow the term “blindspots” from psychology literature [27], wherein it is used to refer to scenarios where an agent makes biased decisions despite exposure to observations that contradicts their decision’s rationale.



Figure 1: $\delta(k)$ measures the relative prevalence of concept c_k in generated images compared with its frequency in natural data. We classify concepts with $\delta(k) < 0.1$ as **suppressed conceptual blindspots** and concepts with $\delta(k) > 0.9$ as **exaggerated conceptual blindspots**. The images generated by four popular generative image models illustrate both blindspot regimes and comparatively aligned concepts.

generative models under-produce or over-produce a broad set of concepts relative to the natural data distribution.

Existing evaluation protocols do not directly answer this question for generative image models [28, 29]. Widely used metrics such as FID [30] summarize image realism or distributional proximity at a coarse level, and therefore can obscure failures concentrated in particular concepts. Related measures based on CLIPScore, generation diversity, anomaly detection, or distribution coverage [31–33] offer useful aggregate information but do not operate at the granularity of fine-grained visual features or concepts [34, 35]. Conversely, qualitative evaluations such as participant studies [5, 17, 36–38] and open-ended model exploration [39] can reveal concrete failures, but they do not provide a scalable, repeatable, and model-comparable procedure for discovering conceptual blindspots.

This work. We develop an automated and unsupervised methodology for identifying concepts whose prevalence differs between the natural data-generating process and a trained generative image model. The paper makes three contributions that connect the formal problem, the technical estimator, and the empirical analysis:

- **Formalizing Conceptual Blindspots in Generative Image Models.** We define conceptual blindspots as concept-level discrepancies between natural images and generated images, yielding a quantitative framework for measuring suppression and exaggeration relative to the data-generating process (Section 3). This moves the evaluation target from anecdotal failures or hand-selected concepts to a principled distributional comparison.
- **A Scalable, Unsupervised Approach for Identifying Conceptual Blindspots using Sparse Autoencoders.** We instantiate the formalization with sparse autoencoders (SAEs), which provide a shared interpretable concept basis for comparing natural and synthesized images (Section 4). We train and open-source an archetypal SAE (RA-SAE) on DINOv2 features with 32,000 concepts, the largest such RA-SAE to date.
- **Exploratory Tool and Analysis.** We provide an exploratory web tool for distribution- and datapoint-level analysis of blindspots across models (Sections 5.1–5.4). Applying the method to Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5), we identify suppressed conceptual blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated conceptual blindspots (e.g., wood background texture and palm trees), as illustrated in Fig. 1.

The model, code, and web tool are available at a software repository.

2 Related Work

2.1 Explainability in Vision

Early explainable-AI work in computer vision emphasized attribution methods that localize influential input regions [40–42]. Although valuable for saliency analysis, these methods provide limited semantic access to learned representations and can produce unreliable explanations [43–45]. Concept-based interpretability [46] addresses this limitation by identifying semantically meaningful directions in neural networks, thereby exposing not only where a model attends but also which concepts and visual structures it uses [47–49].

Recent analyses show that prominent concept-based methods—ACE [44], CRAFT [48], and Sparse Autoencoders (SAEs) [50, 51]—can be understood as variants of a dictionary-learning problem with different constraints [52]. Within this family, SAEs provide a particularly scalable mechanism for extracting sparse, interpretable concept activations. Although standard SAEs have known limitations, including overly specific features [53], compositionality challenges [54], and limited intervention effects [55], improved variants such as archetypal SAE (RA-SAE) [56] are designed to yield more stable and data-faithful dictionaries.

Other interpretability tools, including prompt-based probing [57], attention or activation visualizations [58, 59], and dataset-level statistics [32, 33], provide complementary but partial evidence. They generally operate on full images, prompts, or aggregate coverage measures rather than on fine-grained features and concepts [34, 35]. They also often require subjective interpretation and do not separate distinct failure modes [60]. These limitations motivate an explicitly concept-level diagnostic for generative image models [28].

2.2 Generative Image Models

Diffusion-based methods now dominate generative vision modeling across images [4, 61–64], video [65–72], and 3D generation [7, 73–75]. In image generation, this trajectory begins with denoising diffusion probabilistic models (DDPMs) [76] and extends through non-Markovian denoising diffusion implicit models (DDIMs) [77].

The Stable Diffusion (SD) [9] family made diffusion models broadly accessible to research and open-source communities. The original SD model was followed by SD 2 [78], SD 3 [79], and SD XL [80], together with extensions that condition generation on style [81, 82], pose [83], identity [84, 85], and alternative input modalities such as image-to-image generation. Additional latent diffusion systems, including Kandinsky [86], PixArt [87], DeepFloyd [4, 88], and FLUX [89], expand the set of architectures for which concept-level comparisons are relevant.

2.3 Datasets for Generative Image Models

The performance of contemporary generative vision models depends substantially on computational scale and large internet-derived datasets [90–92]. LAION-5B [93], collected from Common Crawl [94], has been especially influential for open-source text-to-image systems, including SD and its derivatives; it contains more than 5B image-caption pairs, of which 2.3B are in English. Other widely used resources include COYO-700M [95] and Conceptual Captions [96], with 700M and 3M image-caption pairs, respectively.

The adoption of LAION-5B also intensified concerns about bias and unsafe content [97–101]. Despite filtering, harmful material remained [97–99], including NSFW images [100] and hundreds of CSAM instances [101], prompting temporary removal from official channels. LAION-5B further contains low-quality images [102] and internet-style captions, such as product descriptions, that can differ from user prompts supplied to trained models [103]. These properties make concept-level comparison between natural and generated images especially important: aggregate quality metrics may miss systematic distortions inherited from, or amplified relative to, the training distribution.

2.4 Concept Discovery and Sparse Coding in Generative Image Models

Dictionary learning constructs sparse representations in which each sample is reconstructed from a small number of dictionary atoms [104–106]. Building on compressed sensing theory [107, 108], the area has progressed from vector quantization [109] to methods such as Non-negative Matrix Factorization [110, 111], Sparse PCA [112], and K-SVD [113]. More recent developments include online learning [114], structured sparsity [115], theoretical recovery guarantees [116, 117], and increasingly direct connections to deep representation learning [118, 119].

Sparse coding has also been used to study the emergence of high-level concepts inside diffusion models [120]. Before diffusion models, concept-grounded interpretability for generative systems relied on concept-bottleneck models, which require human intervention during training [121], or post-hoc detectors that retrofit concept supervision [122]. Because both strategies depend on human-defined concepts, they can miss systematic concept-level trends that are not specified in advance.

Aggregate metrics, including precision, recall, density, and coverage [35, 123], as well as latent density scores that estimate sample quality from the model’s latent space [124], are useful for measuring global generative behavior. They are less suited to identifying which specific concepts are under- or over-represented. This gap motivates a scalable, unsupervised framework that quantifies concept-level failure modes in generative image models [125].

3 Formalizing Conceptual Blindspots in Generative Models

We formalize *conceptual blindspots* as systematic discrepancies between the concepts expressed in natural images and those expressed in model-generated images. The formalization yields quantitative measures of concepts that are suppressed or exaggerated by a generative model relative to the data distribution. The construction is summarized in Fig. 2. Although the definitions rely on standard assumptions from representation learning and identifiability [126–130], our empirical analysis shows that the resulting measurements expose meaningful model behavior even when these assumptions are only approximate.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(c) = \exp(-E(c))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(c) = \sum_k E(c_k)$ and hence $p(c) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G: \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $c \in \mathcal{C}$ to observations $x \in \mathcal{X}$, i.e., $x = G(c)$.

For notational simplicity, we use $p(\cdot)$ to denote both the latent density $p(c)$ and its push-forward to image space $p(x)$, where $x = G(c)$. This is justified by the invertibility of G , which induces a valid distribution over $\mathcal{X}$ via the change of variables formula. The coordinates of the latent space therefore represent the **Concepts** that structure the image distribution $P_{\mathcal{X}}$ over the observation space $\mathcal{X}$. Such latents may encode attributes including color, shape, size, and location [131, 132]. We also allow the data-generating process to associate each image with a text description $t \in \mathcal{T}$, while leaving the captioning mechanism implicit. A text-conditioned **Generative image model** g_{θ} with parameters θ is then trained on image–text pairs to map a noise signal $\eta \sim \mathcal{N}(0, I)$ and a scene description t to an image x depicting that description.

To identify blindspots in g_{θ} , we compare the probability mass assigned to each concept by the data-generating process with the corresponding mass induced by the model. We begin from an evaluation set of natural image–text pairs $(\mathcal{D}_{\mathcal{X}}, \mathcal{D}_{\mathcal{T}})$ and construct a synthetic set $\mathcal{D}'_{\mathcal{X}}$ by generating images from the same text descriptions. Concept occurrence is estimated with an **Energy model** $\xi: \mathcal{X} \rightarrow \mathbb{R}^d$ that maps an image to a d -dimensional vector whose k^{th} coordinate represents the energy of concept k , i.e., $\xi_k(x) = E(c_k)$. The invertibility assumption on the data-generating process makes these estimates well defined. The probability associated with an image x can then be written as $p_k(x) \propto \exp(-\sum_k \xi_k(x))$ (where Z_{ξ} is the partition function), yielding the population-level estimate $p_k(\mathcal{D}_{\mathcal{X}}) = \prod_{x \in \mathcal{D}_{\mathcal{X}}} p_k(x)$. Using this construction and the sigmoid map $\sigma(\cdot)$, we define the average energy difference for concept k between $\mathcal{D}_{\mathcal{X}}$ and $\mathcal{D}'_{\mathcal{X}}$.

Definition 2 (Energy Difference). Let $x \in \mathcal{D}_{\mathcal{X}}$ denote a real image sampled from the data-generating process G , and let $x' \in \mathcal{D}'_{\mathcal{X}}$ be a synthetic image generated by the model g_{θ} . Let $\xi_k: \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the

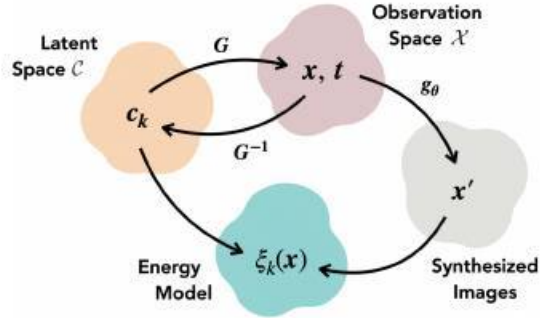


Figure 2: **Intuitive mapping of our framework.** Latent concepts $c_k \in \mathcal{C}$ are mapped to observations (x, t) through the (invertible) DGP. g_{θ} generates images x' based on t . The energy model $\xi_k(x)$ extracts concept representations from both x and x' , enabling comparison of concept distributions to identify blindspots.

energy difference for concept k as:

$$\begin{aligned}\delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{\mathbf{x}'}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x}}[\xi_k(\mathbf{x})]) \\ &= \frac{p_k(\mathcal{D}'_{\mathcal{X}})}{p_k(\mathcal{D}_{\mathcal{X}}) + p_k(\mathcal{D}'_{\mathcal{X}})},\end{aligned}\tag{1}$$

where the expectations are taken over $\mathcal{D}'_{\mathcal{X}}$ and $\mathcal{D}_{\mathcal{X}}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{\mathbf{x} \in \mathcal{D}} \xi_k(\mathbf{x}))$ denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

Thus, the energy difference for concept k expresses the relative probability of observing that concept in a target set of observations (here, $\mathcal{D}'_{\mathcal{X}}$) compared with a baseline set (here, $\mathcal{D}_{\mathcal{X}}$). This statistic directly induces the following definition of conceptual blindspots.

Definition 3 (Suppressed / Exaggerated Conceptual Blindspots). Given a generative image model g_θ , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in the model if $\delta_{g_\theta \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_\theta \leftrightarrow G}(k) > \lambda_{\max}$.

A conceptual blindspot is therefore a concept whose likelihood in generated images deviates markedly from its prevalence under the data-generating process, either through suppression or exaggeration. Suppressed concepts have disproportionately low activation (e.g., $\delta(k) < \lambda_{\min}$), whereas exaggerated concepts are overrepresented (e.g., $\delta(k) > \lambda_{\max}$). Throughout the empirical analysis, we use $\lambda_{\min} = 0.1$ and $\lambda_{\max} = 0.9$ to isolate the two regimes.

This definition is related to “mode collapse” in prior work (e.g., see Bau et al. [39]), but it shifts the unit of analysis. Mode collapse concerns the exaggerated or suppressed probability of generating entire images; conceptual blindspots concern altered probabilities of specific concepts within images. For instance, if a model fails to generate objects with a white background, the missing background attribute is a suppressed conceptual blindspot.

4 Method: Operationalizing the Definition of Conceptual Blindspots

We now instantiate the formal definition as a concrete pipeline for detecting conceptual blindspots in a generative model g_θ . Sec. 3 requires two empirical objects: (i) generated images sampled from g_θ under captions that permit comparison with the natural data-generating process, and (ii) an energy model that maps both natural and generated images into a common concept basis. We use $\|\cdot\|_F$ for the Frobenius norm and $\|\cdot\|_0$ for the number of nonzero entries (the ℓ_0 pseudo-norm). For a vector or matrix $\mathbf{X}$, $\mathbf{X} \geq 0$ denotes element-wise non-negativity. For $n > 0$, $[n] := \{1, \dots, n\}$, and the i -th row of a matrix $\mathbf{A}$ is denoted by $\mathbf{A}_i$.

From Prompts to Latent Representations. To identify conceptual blindspots in g_θ , we compare natural image-caption pairs $(\mathbf{x}, t)$ sampled from the data-generating process G with synthetic counterparts generated by g_θ from the same text descriptions.

Given a caption t , we synthesize the paired image $\mathbf{x}'$ with a pretrained text-to-image generator $g_\theta : \mathcal{T} \rightarrow \mathcal{X}$ implemented as a denoising diffusion probabilistic model (DDPM) [78] [86] [87]. Sampling proceeds in latent space through the reverse trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ follow the standard cosine noise schedule. The final latent γ_0 is decoded by a pretrained VAE to obtain $\mathbf{x}' = \text{VAE}(\gamma_0)$. In the subsequent analysis, g_θ is treated as a black-box prompt-to-image map $t \mapsto \mathbf{x}'$.

Defining the Energy Model. Prior work shows that self-supervised learning can recover structure related to the data-generating process and identify the underlying energy function up to linear transformations [126] [128] [133] [134]. We therefore use DINOv2 [135], a state-of-the-art self-supervised vision model, as the feature extractor for our energy model. Because the number of concepts underlying the DGP may exceed the dimensionality of the model feature space [51] [136], we train sparse autoencoders (SAEs) on DINOv2 features to recover subspaces associated with these concepts [50] [56] [137] [139]. The operational assumption is that if DINOv2 represents concepts through approximately orthogonal directions, as reflected by the independence constraint in Def. 1.

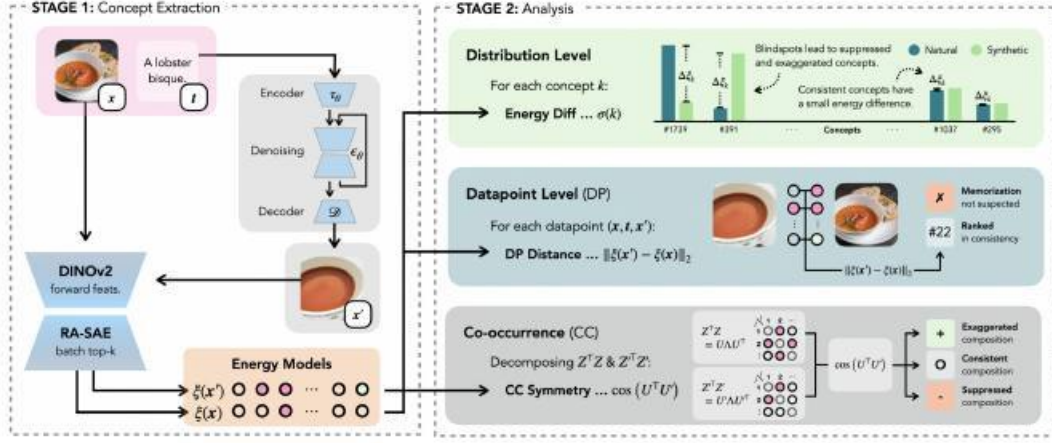


Figure 3: **Concept Extraction Pipeline.** For a triplet (x, t, x') , the concepts in x and t are extracted by obtaining each image’s DINOv2 features, which are further processed by a RA-SAE into sparse concept embeddings, yielding energy models $\xi(x)$ and $\xi(x')$, respectively. In particular, $\xi_k(x) = E(c_k)$ holds the energy in the k -th concept. The activation associated by the SAE to a dimension will serve as our approximation of the ground-truth energy function assigned to the concept modeled by that dimension.

Formally, let $f : \mathcal{X} \rightarrow \mathbb{R}^d$ denote the DINOv2 feature extractor. For each natural or synthetic image x in $\mathcal{D}_\mathcal{X}$ or $\mathcal{D}'_\mathcal{X}$, we compute $a = f(x) \in \mathbb{R}^d$. For n image–text pairs, the natural and generated features are stacked into matrices $A, A' \in \mathbb{R}^{n \times d}$. Each feature vector is then decomposed into a sparse combination of learned concept atoms using an SAE. Let $D \in \mathbb{R}^{d \times K'}$ be a dictionary of K' concept vectors, and let $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ be the encoder that maps features to sparse concept codes. Applying Ψ row-wise produces the activation matrix $Z = \Psi(A) \in \mathbb{R}^{n \times K'}$, with each row $z_i = \Psi(a_i)$ representing the concept decomposition of one image. The decoder reconstructs features as $ZD^\top$, and the SAE objective minimizes reconstruction error subject to sparsity and non-negativity:

$$\min_{\Psi, D} \|A - \Psi(A)D^\top\|_F^2 \quad \text{s.t.} \quad \Psi(A) \geq 0, \|\Psi(A)_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

Standard SAEs can converge to arbitrarily oriented dictionaries, which makes downstream concept analyses sensitive to random initialization. To obtain a reproducible and data-grounded concept basis, we use the *Archetypal* SAE (RA-SAE) [56] with a TOP-K sparsity constraint [137]. RA-SAE constrains the dictionary D to be a convex combination of training examples. Specifically, $D = WA$ with $W \in \Omega_{K', n}$, where $\Omega_{K', n}$ is the set of row-stochastic matrices in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K', n} := \{W \in \mathbb{R}^{K' \times n} \mid W \geq 0, W\mathbf{1} = \mathbf{1}\}. \quad (3)$$

Consequently, every atom D_i lies in the convex hull of the data $\text{conv}(A)$, and every reconstruction $ZD^\top$ lies in the conic hull of the data $\text{cone}(A)$. This constraint keeps learned concepts anchored to the support of the data distribution [56]. After training, the SAE yields aligned sparse codes Z for natural images and Z' for their generated counterparts. *These codes capture the same prompt-conditioned visual semantics in terms of shared, interpretable concepts, with the activation value of the concept serving as energy values for our analysis of conceptual blindspots.* Thus, for each prompt and associated natural image (t, x) , the pipeline synthesizes x' and maps both images into a shared sparse concept space, producing (z, z') for direct comparison of natural and generated visual content.

This pipeline operationalizes conceptual fidelity: a captioned image (t, x) and its generated counterpart x' are encoded by DINOv2 and RA-SAE into interpretable concept representations (z, z') , which form the empirical basis for measuring concept shifts induced by the generative process.

5 Results

We evaluate four generative image models trained on LAION-5B or related derivatives—SD 1.5, SD 2.1, PixArt, and Kandinsky—using $|\mathcal{X}| = 10,000$

image-text pairs and their corresponding generations (Fig. 4). The analysis spans three complementary levels (Fig. 3): a **distribution-level** evaluation identifies suppressed and exaggerated concepts; a **datapoint-level** analysis localizes failures associated with ambiguity, omission, and memorization; and a **compositional** analysis characterizes distortions in concept co-occurrence geometry.

The empirical study is supported by an interactive exploratory tool, shown in Fig. 5. The web interface centers on UMAP projections of concept representations, enabling direct visualization and comparison of concept-level energy differences. It is publicly available at a software repository, together with pre-computed energy-difference data for the four evaluated models (SD 1.5, SD 2.1, PixArt, and Kandinsky). The analyses below use this interface to move between global patterns and individual examples. Its primary functionalities are:

- **Contrast different models and architectures.** For each evaluated model, the tool provides a UMAP visualization over all 32,000 RA-SAE concepts. Each point is one concept, color-coded by its energy difference.
- **Inspect concepts.** Each concept card reports key statistics, representative real and generated images (x, x'), and visualized co-occurrence patterns.
- **Explore blindspots.** Beyond UMAP and per-concept views, the tool provides global rankings of suppressed and exaggerated blindspots, making the most pronounced concept-level deviations easy to inspect.

Rather than exhaustively analyzing a single failure mode, we use the tool to organize a set of high-level findings that demonstrate how the framework supports customizable concept-level investigation.



Figure 5: **Overview of the Exploratory Tool.** The web interface presents a UMAP projection for each evaluated model, with each dot corresponding to one concept and color indicating its energy difference. Selecting a concept opens a detail panel containing illustrative images, statistics, representative natural and generated images x and x' , an ordered list of co-occurrences, and global rankings of blindspots.

5.1 Generative Image Models Cannot Produce Unseen Concepts

To measure concept-level disparities between generated and natural images, we first analyze the marginal energy difference $\delta(k)$ for the 32,000 RA-SAE concepts. As defined in Sec. 4 $\delta(k)$ mea-

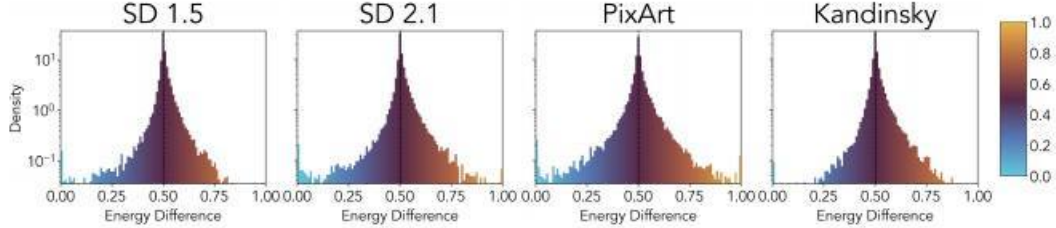


Figure 6: **Concept Energy Distribution.** Log-scale histograms of energy differences $\delta(k)$ over 32,000 concepts for each evaluated model, comparing concept prevalence in natural and synthesized images. Values left of zero indicate suppressed concepts; values right of zero indicate exaggerated concepts.

282 sures the relative prevalence of concept k in synthesized images compared with natural images.
 283 Values $\delta(k) < 0.1$ indicate under-representation, or suppression, whereas values $\delta(k) > 0.9$ indicate
 284 over-representation, or exaggeration. Fig. 6 reports the resulting distributions for the four evaluated
 285 models. All models exhibit heavy-tailed histograms with substantial mass near both extremes, demon-
 286 strating that concept coverage errors are not confined to isolated cases. The left tail, corresponding
 287 to suppressed concepts, is consistently denser and longer than the right tail, indicating a stronger
 288 tendency toward omission than exaggeration. This asymmetry is quantified by negative skewness
 289 in every model: Skewness = -0.54 for SD 2.1, -0.40 for both SD 1.5 and PixArt, and -0.23 for
 290 Kandinsky.

291 Although all four models contain both suppressed and exaggerated concepts, their profiles are not
 292 identical. PixArt, for example, exhibits a wider spread, consistent with a more dispersed suppressed-
 293 concept distribution. The shared left-skew establishes concept omission as a recurring property across
 294 the evaluated models, while the model-specific distributional shapes motivate the structural analysis
 295 in the following sections.

296 5.2 Structure and Specificity of Conceptual Blindspots

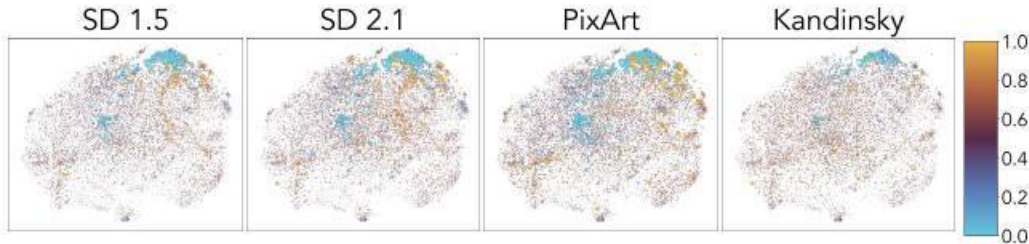


Figure 7: **Structure of Concept Energy Differences.** UMAP visualizations of 32,000 concepts, colored by energy difference $\delta(k)$ between natural and synthesized images. Clustered regions reveal structured conceptual blindspots, with suppressed concepts shown in blue and exaggerated concepts in yellow.

297 Having established marginal concept-frequency discrepancies, we next examine whether blindspots
 298 are organized in concept space. We embed all 32,000 concepts into two dimensions with UMAP
 299 applied to the sparse codes and color each concept by $\delta(\cdot)$. Fig. 7 shows that blindspots form distinct
 300 clusters across all models. These clusters often correspond to contiguous regions of suppressed
 301 concepts, indicating that blindspots are structured rather than randomly distributed and may reflect
 302 shared properties of training data or architectural priors. We then quantify how consistently these
 303 structures appear across models by comparing concept-level $\delta(\cdot)$ vectors. Fig. 8 reports scatter plots
 304 and pairwise Pearson correlations between SD 1.5 and the other models. SD 1.5 and SD 2.1 are
 305 strongly correlated ($r = 0.82$), consistent with their related architectures and training pipelines.
 306 Correlations with PixArt and Kandinsky are lower— $r = 0.41$ and $r = 0.46$, respectively—showing
 307 that different model families emphasize different regions of the concept space.

308 Overall, these results show that conceptual blindspots include both shared and model-specific compo-
 309 nents. Shared blindspots plausibly reflect properties of the data distribution, whereas model-specific

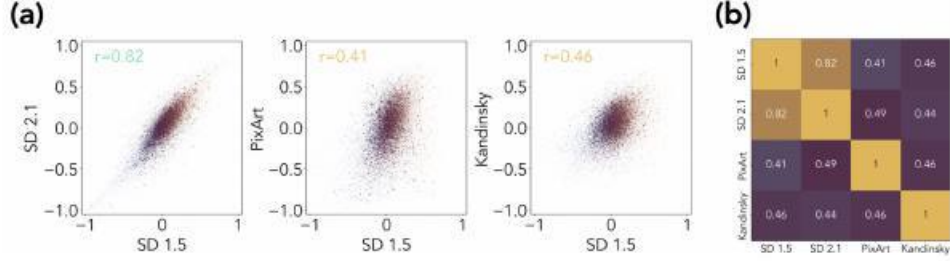


Figure 8: **Cross-Model Concept Energy Correlation.** Pairwise scatter plots of $\delta(k)$ across all four evaluated models, with Pearson correlation coefficients reported in the upper left. Strong alignment between SD 1.5 and SD 2.1 contrasts with weaker correlations involving other architectures, indicating model-specific blindspot profiles. **Correlation Matrix of Conceptual Blindspots.** Heatmap of pairwise Pearson correlation coefficients for δ between all models, quantifying how much conceptual blindspot structure is shared across models.

blindspots point to differences in architecture, training dynamics, or capacity; the qualitative analysis below therefore examines both universal and unique cases.

5.3 Qualitative Examples of Blindspots

We next visualize specific examples of both suppressed and exaggerated blindspots to gauge what concepts fall under these regimes. Specifically, in Fig. 9a we show an example of a conceptual blindspot suppressed by all models—we find all evaluated models fail to reproduce the concept `solid white on documents`. As can be seen in the figures, despite the caption explicitly referencing this concept, none of the generated images reflect the intended visual semantics, suggesting that this region of the concept space is systematically under-sampled across models. Meanwhile, Fig. 9b highlights a model-specific blindspot: the concept `pan` is accurately captured by three models, yet conspicuously missing from generations produced by Kandinsky. This reinforces the findings from Sec. 5.2 where cross-model agreement was found to be high in some cases but limited in others.

Conversely, in Fig. 10 we present a case of exaggeration, where the concept `shadow under animal` is overly emphasized in generated images. While shadows are mildly plausible, their consistent and pronounced rendering across models, relative to the more nuanced and variable occurrences in natural images, suggests an overactive prior. Interestingly, despite attempts at finding concepts that are uniquely exaggerated by a specific model, we did not find any clear examples—this suggests exaggerations are approximately universal.

Overall, the examples above concretely demonstrate how conceptual blindspots manifest in generated outputs, illustrating that our energy-based diagnostic can surface both shared and model-specific failure modes. Notably, it enables the identification of surprising model

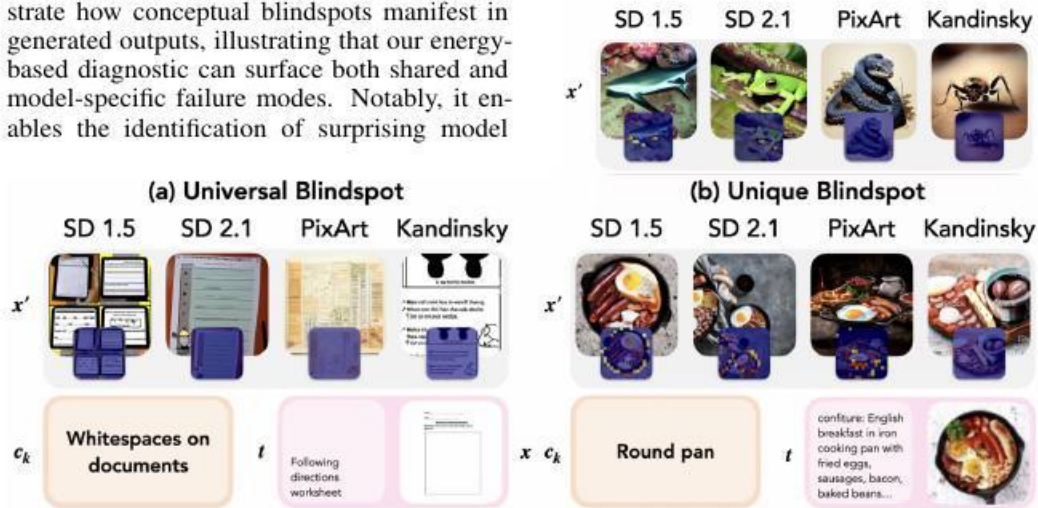


Figure 9: **Examples of Suppressed Conceptual Blindspots.** The energy-based diagnostic can surface both shared and model-specific failure modes. The universal blindspot is present in all evaluated models; the unique blindspot is only present in Kandinsky.

334 limitations—such as the consistent failure to re-
 335 produce clear or solid background elements, like
 336 whitespaces on documents, across all mod-
 337 els. This raises the possibility that certain failure
 338 patterns may stem from architectural constraints
 339 or training data biases that transcend individual
 340 model idiosyncrasies.

341 While these aggregate-level analyses are
 342 informative, they invite a deeper ques-
 343 tion: do these blindspots emerge only in
 344 the aggregate across many samples, or
 345 do they manifest themselves even at the
 346 level of individual datapoints? This finer-
 347 grained perspective allows us to probe
 348 the mechanisms behind blindspots more
 349 directly—uncovering cases of prompt mis-
 350 interpretation, latent memorization, or
 351 both.

353 5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

354 To move beyond population-level statistics, we examine individual natural vs. generated image pairs
 355 for which the $\delta(\cdot)$ values averaged across all concepts exhibit the largest and smallest differences.
 356 This analysis aids easy understanding of model
 357 success and failures, latter of which we find
 358 often arises from prompt ambiguity or memo-
 359 rization artifacts. For example, Fig. 11a shows
 360 instances with near-zero difference in average
 361 $\delta(\cdot)$ values. In these cases, the generated im-
 362 ages are conceptually indistinguishable from the
 363 original. However, qualitative inspection clearly
 364 shows this happen not because the model faith-
 365 fully captures the prompt semantics, but from
 366 pure replication of memorized templates: we
 367 see repetitive visual structures (e.g., outlines
 368 of clothing or object arrangements), indicating
 369 that the model may be copying from overly fre-
 370 quent patterns in the training data. By contrast,
 371 Fig. 11b illustrates samples that are among the
 372 largest $\delta(\cdot)$ values. These indicate significant
 373 conceptual divergence between the synthesized
 374 and natural image. While some of these dis-
 375 crepancies can be attributed to underspecified or
 376 noisy captions, others reveal genuine blindspots:
 377 the prompt describes a clear concept faithfully present in $\mathcal{D}_{\mathcal{X}}$, yet the model fails to realize it in $\mathcal{D}'_{\mathcal{X}}$.
 378 This failure suggests that even when language grounding is adequate, certain concepts fall outside the
 379 model’s generative abilities.



Figure 11: • **Datapoint-level Conceptual Alignment.** (a) Examples with minimal energy differences where models appear to memorize training patterns. (b) Examples with large differences where significant concept divergences occur.

380 5.5 Analyzing Post-Training Effects

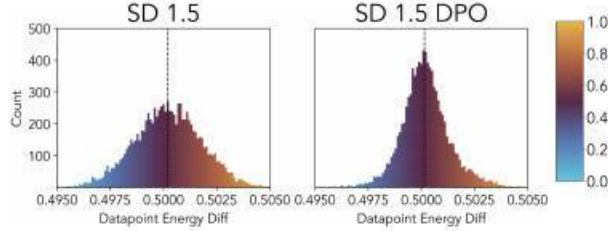


Figure 12: **Effect of DPO on Concept Fidelity.** Histogram of datapoint-wise energy differences between the synthesized and natural distribution of SD 1.5 models with and without DPO.

381 Post-training protocols, e.g., safety fine-tuning, have been argued to reduce the diversity of model
 382 generations [140]. Given our pipeline’s ability to isolate interesting differences in a model’s gen-
 383 erations and the ground-truth DGP, we next use it to understand the effects of DPO—a popular
 384 safety fine-tuning protocol [141]. Specifically, we compare two variants of the SD 1.5 model: one
 385 trained with DPO, and one without. For each image pair $(\mathcal{D}_X, \mathcal{D}'_X)$, we compute the ℓ_2 norm of
 386 the difference between their internal concept energy vectors, $\|\xi(\mathcal{D}'_X) - \xi(\mathcal{D}_X)\|_2$. Fig. 12 presents
 387 a histogram of these datapoint-wise energy differences. The DPO-enhanced model exhibits both a
 388 lower median and a narrower spread, indicating more consistent distribution of generated concepts
 389 with the ground-truth DGP. This suggests that DPO may serve to regularize the model’s concept
 390 distribution, encouraging outputs that more closely reflect the semantic content of the seen inputs.
 391 While our analysis does not disentangle the specific inductive biases introduced by DPO, these results
 392 provide empirical evidence that its optimization objective, which favors human-preferred generations,
 393 indirectly promotes better match with the training distribution. In particular, it reduces *both over and*
 394 *under* activation of individual concepts relative to the natural baseline. These findings highlight the
 395 utility of our pipeline in characterizing the downstream effects of post-training interventions: not
 396 merely in terms of output quality, but in how they reshape the conceptual geometry of the model’s
 397 output space.

398 5.6 Conceptual Misalignment as a Function of Empirical Frequency

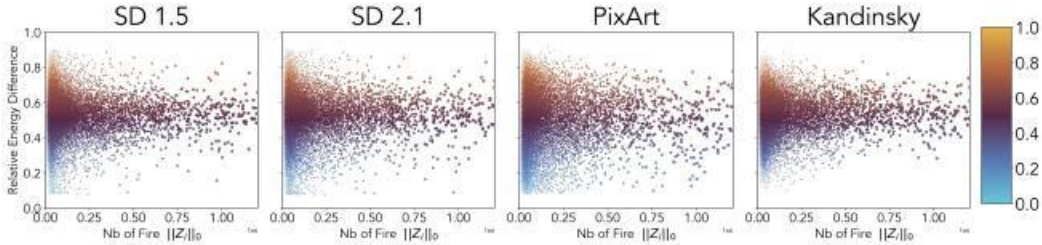


Figure 13: **Concept Fidelity Across Frequency Spectrum.** Scatter plots showing the relationship between concept frequency (x-axis) and the energy difference (y-axis) across four evaluated models. Each point represents a concept with size is proportional to its activation frequency.

399 We previously hypothesized that conceptual blindspots are not merely be architectural artifacts,
 400 but may also emerge as a direct consequence of distributional peculiarities of certain features. In
 401 this section, we empirically test this hypothesis by examining whether concepts that are rarely
 402 activated in natural images, i.e., those in the long tail of the data distribution, lead to blindspots
 403 in generative models. Specifically, we process the natural dataset $\mathcal{D}_X$ through the trained SAE
 404 and compute, for each concept k , its empirical frequency $\|\mathbf{Z}_{:,k}\|_0$, where $\mathbf{Z}_{:,k}$ is the activations of
 405 concept k across all our images. We then correlate this with the absolute energy difference observed
 406 across generated outputs. Fig. 13 visualizes this relationship for all evaluated models. We find
 407 that concepts with higher frequency in natural data tend to show lower energy discrepancies, while
 408 rare concepts—especially suppressed ones ($\delta(k) < 0.5$)—exhibit significant alignment errors. This
 409 suggests that many blindspots stem not from randomness or model quirks, but from systematic effects
 410 tied to long-tail concept distributions. Addressing these issues may require not just architectural
 411 changes but also strategies like data reweighting or augmentation.

6 Discussion

Our analysis reveals multiple conceptual blindspots in four popular generative image models. The results presented here, however, only scratch the surface: each individual finding could warrant its own dedicated investigation. Rather than delving deeply into any one of these questions, we instead showcase the versatility of our method and exploratory tool. Out of the box, they allow for a systematic identification of concepts that models struggle to generate, detection of memorization artifacts, discovery of datapoints with insufficient captions, quantification of post-training effects, and characterization of conceptual shifts across model architectures. We thus open space for follow-up work to extend the depth of analysis, scope of evaluated architectures, and inquiry into root causes of conceptual blindspots. Future work could also explore hierarchical representations of concepts in RA-SAE to allow for a more nuanced analysis.

Limitations. We wish to highlight several limitations of our work. By relying on DINOv2 and RA-SAE for concept extraction and representation, our approach is inherently constrained to the kinds of concepts these models capture; concepts poorly represented by them will escape our analysis. Additionally, while our sample size of 10,000 images is substantial, it may not fully capture the long tail of rare concepts, concept co-occurrence, or other compositional statistics.

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Appendix

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A Experimental Setup

This section details the experimental setup for our analysis of four popular generative image models: SD 1.5/2.1, Kandinsky, and PixArt, all trained on LAION-5B or its subsets/derivatives. The code is available at a software repository.

A.1 Observation Space

The observation space is constructed by sampling 10,000 image-text pairs from the LAION-5B dataset [93], which serves as our domain of natural images. Due to concerns with CSAM and other unsafe content in the dataset, the original data release is no longer available. A substitute release of a subset of this dataset with additional filtering of the unsafe content, available at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe> is used.

The sampling procedure consists of: (1) loading the full LAION dataset using the Hugging Face `datasets` library, (2) performing validation to ensure proper URL structure and resource availability via HTTP HEAD requests, and (3) employing random sampling with replacement until reaching the target count of 10,000 valid samples. This approach, yielding D_G with (x, t) tuples, ensures our observation space contains accessible image-text pairs for comparative analysis of a dataset of image URLs whose large portion has been made unavailable since original release. Additional examples of synthesized images are shown in Appendix H

A.2 Synthesized Images

For each of the four evaluated models, we generate a synthetic dataset D_{g_θ} to have a one-to-one correspondence with D_G , yielding triplets (x, x', t) . Specifically, given the 10,000 image-text pairs (x, t) from D_G , we use t to synthesize counterpart images x' using each generative model g_θ .

The synthesis process follows the standard text-to-image generation pipeline for each model architecture, implemented using the Hugging Face `diffusers` library, where the models are loaded at mixed precision (fp16). All synthetic images are generated at 512×512 pixel resolution with default parameters.

Stable Diffusion 1.5. The checkpoint from <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5> (which is a mirror of the deprecated <https://huggingface.co/runwayml/stable-diffusion-v1-5>) is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant of SD 1.5 (used in the analysis in Sec. 5.5) follows the baseline SD 1.5 implementation, but replaces the UNet component with a DPO-trained version from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>

Stable Diffusion 2.1. The checkpoint from <https://huggingface.co/stabilityai/stable-diffusion-2-1> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

Kandinsky. The checkpoint from <https://huggingface.co/kandinsky-community/kandinsky-2-1> is used. Inference is performed using 100 inference steps, with the guidance scale fixed at 4.0.

PixArt. The checkpoint from <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS> is used. Inference is performed using 50 inference steps, with the guidance scale fixed at 7.5.

A.3 Distribution Level Analysis

Section 5.1. We compute energy differences $\delta(\cdot)$ across all 32,000 concepts for each evaluated model. The sigmoid transformation with temperature $T = 0.8$ is applied during normalization. The resulting values are visualized as log-scale density histograms with 100 bins spanning $[0, 1]$.

901 **Section 5.2** We embed the complete set of 32,000 concepts into two-dimensional space using
 902 UMAP applied to the sparse concept codes. Each point in this UMAP represents an individual con-
 903 cept, colored according to its energy difference $\delta(\cdot)$, emphasizing both suppressed and exaggerated
 904 blindspots. To quantify cross-model consistency, we compute pairwise Pearson correlation coeffi-
 905 cients between $\delta(\cdot)$ vectors of all model pairs, producing both scatter plots and correlation matrices.
 906 This analysis reveals whether blindspots cluster in conceptual space and identifies model-specific
 907 versus universal patterns of conceptual blindspots.

908 **Section 5.3** We rank all 32,000 concepts by their energy difference $\delta(\cdot)$, and manually examine the
 909 extrema (both suppressed and exaggerated blindspots). For suppressed blindspots, we select concepts
 910 with $\delta(\cdot) < 0.1$; for exaggerated blindspots, we choose those with $\delta(\cdot) < 0.9$. Presented examples
 911 are manually annotated with textual descriptions of the respective concepts through inspection of
 912 their most activating images and spatial attention patterns. We outline ongoing efforts to automate
 913 this concept interpretation in Appendix D

914 **Section 5.5** We compare 1.5 with and without DPO in the following fashion: for each image
 915 pair (x, x') , we compute the L2 norm of the difference between their concept energy vectors
 916 $\|\xi(x') - \xi(x)\|_2$. We apply a sigmoid transformation with temperature $T = 0.8$ to the element-wise
 917 differences before taking their mean. This yields datapoint-wise energy differences that quantify how
 918 much each generated image deviates from its natural counterpart in concept space. Finally, these
 919 differences are visualized as overlapping histograms, contrasting both model variants.

920 **Section 5.6** For each concept c_k , its empirical frequency $\|\mathbf{Z}_{:,i}\|_0$ (the count of non-zero activations
 921 across the natural dataset) is counted. A sigmoid normalization with temperature $T = 0.4$ is then
 922 applied to the energy differences $\delta(\cdot)$. The analysis is visualized using scatter plots where the x-axis is
 923 the empirical concept frequency and the y-axis is the sigmoid-transformed energy difference. The
 924 point sizes are proportional to activation frequency and point colors are proportional to the magnitude
 925 of energy differences.

926 A.4 • Datapoint Level Analysis

927 **Section 5.4** For each image pair (x, x') , we compute the L2 norm of the difference between their
 928 concept energy vectors $\|\xi(x') - \xi(x)\|_2$. This yields a scalar measure of conceptual divergence for
 929 each image pair. The samples are ranked by their energy differences. Minimal divergence indicate
 930 potential memorization artifacts and maximal divergence point to significant conceptual failures. This
 931 analysis enables qualitative inspection of specific failure modes.

932 A.5 • Co-occurrence Analysis

933 **Appendix F** For both the natural and synthesized data D_G and $D_{g\theta}$, concept co-occurrence
 934 patterns are analyzed through the co-activation matrix $Z^T Z$, which holds pair-wise correlations
 935 in concept usage. Spectral analysis is performed to examine the dominant conceptual directions
 936 using eigendecomposition. The alignment between natural and synthetic co-occurrence structures is
 937 assessed using cosine similarity heatmaps between the top-100 eigenvectors of each co-occurrence
 938 matrix. These 100×100 similarity matrices are visualized as square heatmaps where perfect diagonal
 939 alignment would indicate identical principal concept axes, while off-diagonal patterns reveal would
 940 revolve rotations and mismatches in compositional geometry.

B Custom RA-SAE Training Details

This section summarizes the training details of our custom archetypal SAE (RA-SAE) [56]. The model is open-sourced at a software repository.

Dataset. The auto-encoder is trained on the complete ImageNet-1k training split, (≈ 1.28 M) RGB images. Each image is converted to 261 visual tokens using DINOv2 [135]; tokens are fed to the SAE without class or position embeddings. The total number of training tokens is therefore $50 \times 1.28\text{M} \times 261 \approx 1.67 \times 10^{10}$.

Dictionary. The dictionary has 32,000 concept dimensions. For the sparse activation rule, top- k masking with $k = 5$, is used; activations outside the largest five per input are set to 0. The weights are initialized using Xavier/Glorot. The training is conducted at mixed precision (fp16), with the last ten epochs performed at full precision.

Optimizer and Schedule. The model is trained for 50 epochs using base AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$) optimizer is employed with weight decay set to 10^{-5} . Linear warm-up is applied on the first 5% of steps, followed by cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. MSE loss is used alongside an auxiliary term penalizing activations that never enter the top- k set, where $\lambda = 10^{-5}$.

C Computational Resources

This section summarizes the GPU resources used for training and experiments in support of this paper. In total, we used approximately 202 GPU-hours on NVIDIA H100s and H200s.

RA-SAE. Trained for approximately 24 GPU-hours on three NVIDIA H100s.

Synthesized Images. Generating the full $D_{g\theta}$ (see Appendix A) took roughly 5 hours per generator when distributed across four NVIDIA H200 GPUs. With five generators, this totaled approximately 100 GPU-hours on a NVIDIA H200.

● **Distribution-Level Analysis.** Extracting energy differences at the distribution level took about 3 hours per generator on a single NVIDIA H200 GPU (total ~ 15 GPU-hours).

● **Datapoint-Level Analysis.** Computing datapoint-level energy differences, ranking concepts per datapoint, and ranking datapoints per concept also took approximately 3 hours per generator on one H200 GPU (total ~ 15 GPU-hours).

D Autointerpretability

In an effort to automate the interpretation of concepts identified by the RA-SAE, we qualitatively evaluated the ability of Vision-Language Models (VLMs) to describe them. In particular, we queried ChatGPT 4o via the OpenAI API using various prompts and formats of an image x'' in which the given concept was salient, recording the description generated by the VLM. In this section, we highlight the findings from our exploration. For evaluation purposes, we allowed the VLM to provide its full reasoning; to make this a scalable solution, a simple modification of the prompt—asking the VLM to provide only the description—would filter out the reasoning.

First, we provided the VLM with two images: a raw version of x'' and x'' with a heatmap superimposed to localize the intensity of the concept:

Input



You are given two images: (1) raw image and (2) the same image with a heatmap of areas that are salient for a concept present in the image, as identified by a DINOv2-based SAE. Describe what is this concept.

Output

The concept identified by the DINOv2-based SAE in these images appears to correspond to **"ring"** or **"metal ring"** as the salient object.

Reasoning:

- In the original image, a sleek, brushed-metal ring is centered on a wooden surface.
- In the heatmap overlay, the regions that are most activated (least transparent with bright color blocks) are tightly focused on the ring itself, particularly the outer surface and inner reflections.
- The wooden background is mostly ignored except for minor scattered activations, confirming that the concept is not about the background texture.
- The consistent and strong focus on the metallic, circular object suggests that the model recognizes this as the defining element—likely trained to detect the shape, material, and reflectivity typical of rings.

As seen in the example, the VLM did not interpret the heatmap visualization correctly. Instead of describing the wood texture in the background—the actual salient region—it described the ring in the foreground, which received no attention from the RA-SAE.

To determine whether this was a prompting issue, we tested the same scenario again—this time using an expanded prompt that explained the heatmap in greater detail:

Input



You are given two images: (1) raw image and (2) the same image with a superimposed heatmap of areas that contain a concept identified by a DINOv2-based SAE. Red areas are the most salient (contain the concept); blue areas are the least salient (do not contain the concept). Describe what is this concept.

Output

The concept identified by the DINOv2-based SAE in these images appears to be **"ring"** or more specifically **"metal band/jewelry ring."**

Here's the reasoning:

- In the heatmap image, the red and yellow (most salient) regions align very closely with the circular metal ring at the center of the image.
- The ring is the only distinct object in the photo and is clearly segmented from the background (wooden surface), which is mostly marked in blue and green (least salient).
- The attention does not focus on the texture of the background or lighting but is highly localized to the curved, metallic object.

Thus, the concept is almost certainly the metallic ring—its shape, reflectivity, and material seem to trigger the sparse autoencoder's activation, making it a learned concept in the SAE dictionary.

This prompt modification did not resolve the issue, as the VLM still identified the ring as the prominent concept in the image. Several other prompting techniques, including those describing the heatmap in even greater depth (e.g., by including the code used for its generation), were explored with similar results.

987 We next tested providing the VLM only with the heatmap-superimposed x'' and a similar prompt
 988 explaining how to interpret the heatmap:

Input	Output
 You are given an image with a superimposed heatmap of areas that contain a concept identified by a DINOv2-based SAE. Red areas are the most salient (contain the concept); blue areas are the least salient (do not contain the concept). Describe what is this concept.	The concept identified by the DINOv2-based SAE in this image appears to be "ring" or more broadly, "circular metallic object". The most salient (red) areas are focused tightly around the ring itself—especially along its edges and reflective interior—indicating that the model has strongly associated those regions with the concept. The surrounding wooden surface is mostly in blue and green, which shows it is not relevant to the concept, supporting the idea that the model is isolating the ring as the key object of interest.

989 Once again, the result failed to identify the wooden background as the prominent concept in the image
 990 and instead focused on the ring. As in the previous case, we experimented with various prompting
 991 strategies using this image format, but the results remained largely unchanged.

992 From these observations, we concluded that the heatmap alone is insufficient to guide the VLM's
 993 attention to the salient areas. The presence of less relevant regions appears to distract the model. To
 994 address this, we replaced the heatmap-superimposed x'' with an alpha-masked version of x'' :

Input	Output
 You are given an image where only areas that contain a concept identified by a DINOv2-based SAE are preserved; areas without the concept were removed. Describe what is this concept.	The concept identified and preserved in this image appears to be "wood texture" or "wood grain." The retained areas show consistent patterns typical of wooden surfaces—linear, fibrous streaks in earthy tones like beige, gray, and brown. The selective masking highlights parts of the image where this natural texture is most prominent, suggesting the SAE learned to isolate the visual features of wood (likely flooring or a table).

995 As shown in the example, the VLM was now able to correctly identify wood texture as the dominant
 996 concept. We observed consistent success across a range of concepts and images.

997 The formatting of x'' and the prompt shown above yielded the most reliable results in our qualitative
 998 evaluation. However, we note that this evaluation is limited by its qualitative nature (due to the
 999 absence of ground truth annotations) and its focus on a single VLM. We hope future work on the
 1000 autointerpretability of SAE concepts can build on and expand this analysis.

E Additional Results: Model-Specific Blindspots

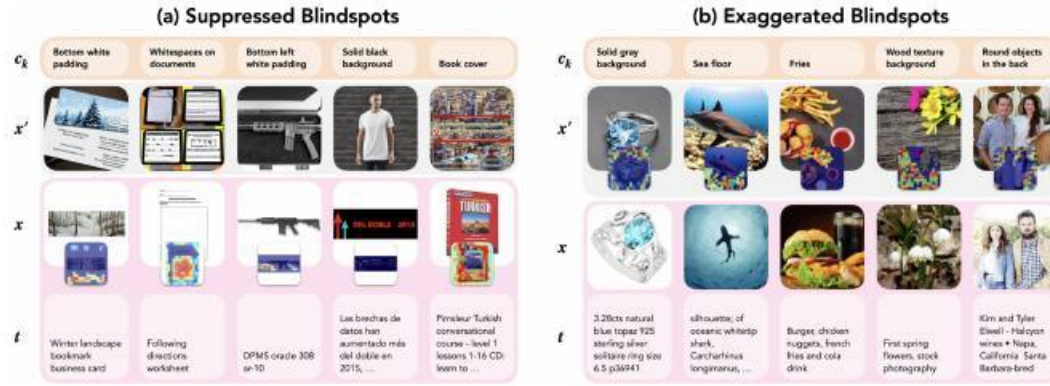


Figure 14: Examples of conceptual blindspots in **Stable Diffusion 1.5**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots), based on the highest absolute activation, is shown. The spatial heatmap for the concept is superimposed atop the image.

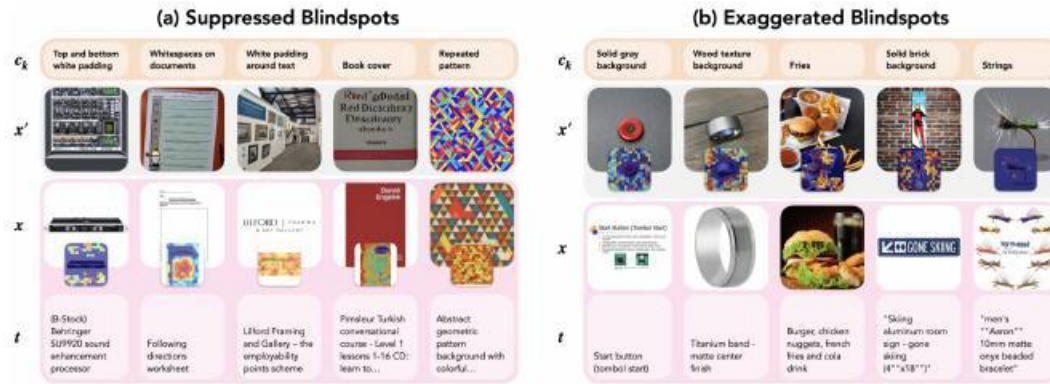


Figure 15: Examples of conceptual blindspots in **Stable Diffusion 2.1**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots), based on the highest absolute activation, is shown. The spatial heatmap for the concept is superimposed atop the image.

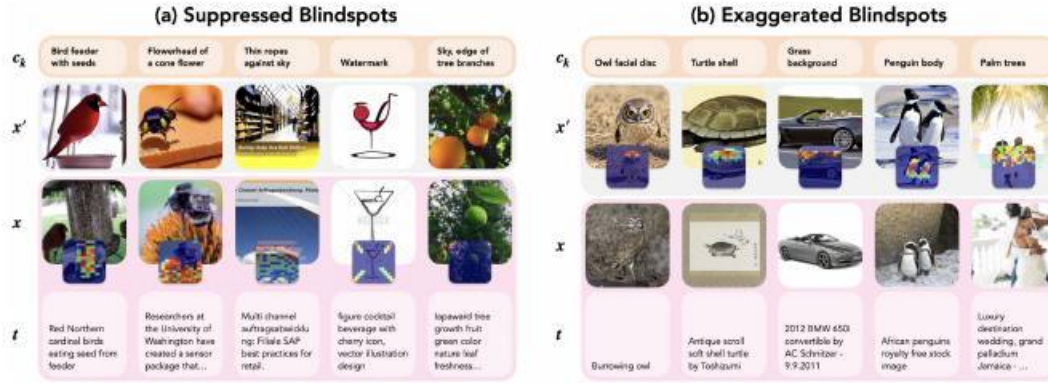


Figure 16: Examples of conceptual blindspots in **Kandinsky**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots), based on the highest absolute activation, is shown. The spatial heatmap for the concept is superimposed atop the image.



Figure 17: Examples of conceptual blindspots in **PixArt**. For each concept, the prototypical natural (for suppressed blindspots) or synthesized (for exaggerated blindspots), based on the highest absolute activation, is shown. The spatial heatmap for the concept is superimposed atop the image.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

Thus far, our analysis has centered on individual concept activations. Yet visual scenes are rarely composed of isolated concepts; instead, they are structured through rich and structured co-occurrence patterns that encode compositional semantics. We now examine whether generative models capture this higher-order structure by analyzing the co-activation matrix $Z^T Z$, which reflects pairwise correlations in concept usage.

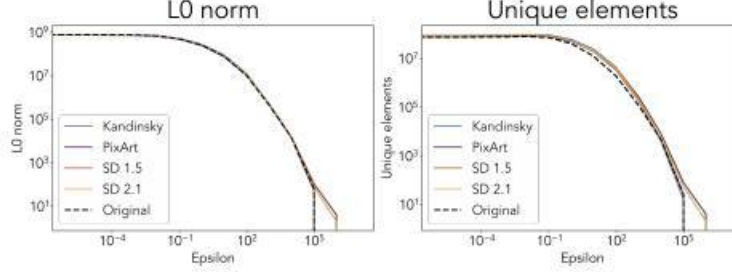


Figure 18: **Sparsity and Structural Divergence.** On the left: L0 norm of the co-occurrence matrix ZZ^T as a function of ϵ (threshold), indicating how many entries remain active in each model. On the right: Number of unique entries in the synthesized distribution relative to the natural distribution. All evaluated models preserve global sparsity structure, but diverge in activation content.

Surprisingly, when assessed at the level of binary structure, diffusion models approximate the global sparsity of the natural co-occurrence matrix with high fidelity. As shown in Fig. 18 (left), the ℓ_0 norm of $Z^T Z$ – thresholded at varying ϵ values – tracks closely between the natural and synthesized distributions across all models. This indicates that the gross connectivity of the conceptual graph, i.e., which concepts tend to co-activate at all, is well preserved. Formally, one can deem $Z^T Z$ as the adjacency matrix of a weighted, undirected graph over concepts, where edge weights reflect co-activation strength across the dataset.

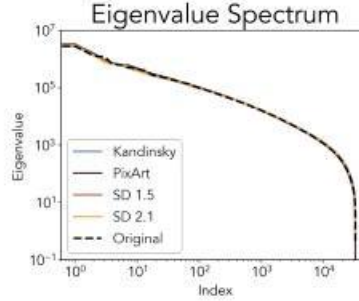


Figure 19: **Spectral Structure of Co-occurrence.** Log-log plot of the eigenvalue spectra from co-occurrence matrices ZZ^T across models and the natural distribution. All evaluated models match the heavy-tailed decay of the natural distribution.

However, as illustrated in Fig. 18 (right), the specific content of these co-activations diverges: a substantial portion of entries in the model-generated $Z^T Z$ are not shared with the natural baseline. This suggests that while the capacity for compositionality is retained, the identity of active pairings may shift, potentially reflecting model specific inductive biases or training artifacts. To probe the internal structure of these co-occurrence patterns, we turn to spectral analysis. Fig. 19 shows the eigenvalue spectra of the co-occurrence matrices for each model and the natural distribution. All spectra exhibit a heavy-tailed decay, consistent with power-law behavior, indicating that generative models preserve the overall rank structure and variance allocation across conceptual dimensions.

Further, we examine the alignment of dominant conceptual directions via cosine similarity heatmaps between the top 100 eigenvectors of the synthesized and natural co-occurrence matrices (Fig. 20). While all models exhibit partial diagonal alignment—implying overlap in principal concept axes—the

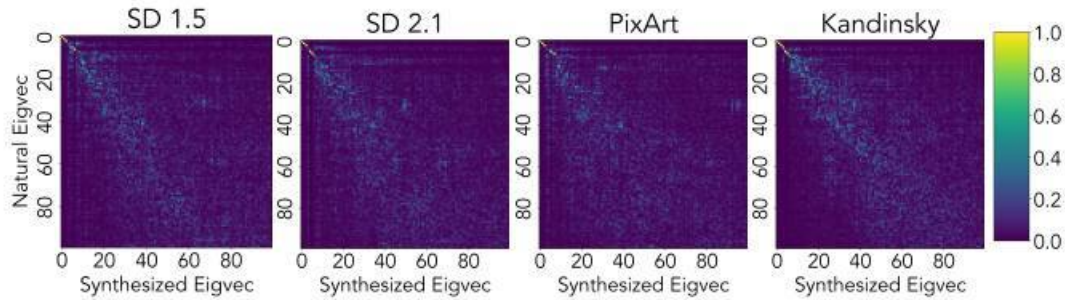


Figure 20: ● **Concept Basis Similarity.** Cosine similarity heatmaps between the top 100 eigenvectors of the natural and synthesized co-occurrence matrices ZZ^T . Diagonal structure shows alignment of dominant conceptual directors, with varying degrees of alignment across the four models.

1028 off-diagonal entries reveal rotations and mismatches in higher modes, reflecting evident deviations in
 1029 compositional geometry.

1030 Together, these findings reveal that diffusion models approximate the global shape of concept co-
 1031 activation surprisingly well, yet deviate in subtle and structured ways when examined through
 1032 the spectral lens. Such higher-order discrepancies may underpin failures in generating coherent,
 1033 multi-object scenes or relational concepts.

1034 G Additional Results: Blindspot Stress Testing

1035 To stress-test the blindspots identified by our method, we gathered a range of prompts describing
 1036 these blindspots and used them to generate many images. We then contrasted the outputs from models
 1037 in which the concept was identified as a blindspot with those in which it was not.

1038 Specifically, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
 1039 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
 1040 *text-to-image model. Return these as a list of strings in Python.* Five images were generated per
 1041 prompt and analyzed using our custom RA-SAE model (see Appendix B), which ranked them by
 1042 the intensity with which the desired concept appeared. All images were then manually reviewed to
 1043 determine whether the blindspot was successfully depicted.

1044 As seen in the following examples of suppressed concepts, while some aspects of the target concept
 1045 occasionally appeared (e.g., a holder or string for the bird feeder blindspot and a round hole for
 1046 the glossy DVD disc blindspot), the models generally failed to generate the full concept. This
 1047 aligns with our method’s assessment and supports the validity of the stress test.

1048 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Examples of images generated with various prompts involving the bird feeder concept as a part of the stress testing. In Kandinsky, our method identified this concept as a suppressed conceptual blindspot, which matches the observed behavior: the model is unable to generate a corresponding image. By contrast, SD 1.5, in which this concept was not identified as a blindspot, is able to generate this concept.



Figure 22: Examples of images generated with various prompts involving the glossy DVD disc concept as a part of the stress testing. In SD 1.5, our method identified this concept as a suppressed conceptual blindspot, which matches the observed behavior: the model is unable to generate a corresponding image. By contrast, Kandinsky, in which this concept was not identified as a blindspot, is able to generate this concept.

1050 **H Additional Examples: Synthesized Images**



Figure 23: Additional image-caption pair examples from LAION-5B with matching images generated with the same prompt by SD 1.5, SD 2.1, PixArt, and Kandinsky.

1051 I Concentration bounds for δ

1052 In our experiments, we estimate $\delta(k)$ using $n = 10,000$ paired samples for each concept. While this
 1053 budget is modest, it raises the natural question of whether it suffices to obtain reliable estimates. To
 1054 address this, we derive a concentration bound on the empirical estimator $\hat{\delta}_n(k)$ using McDiarmid's
 1055 inequality [142]. The resulting bound is tight and demonstrates that even with relatively few samples,
 1056 we can obtain fast and accurate estimates of concept bias.

1057 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1058 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1059 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1060 Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1061 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1062 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1063 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1064 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1065 By McDiarmid's inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1066 which proves the claim. □

1067 Practically, most concept scores $\delta(k)$ are sparse, with the majority concentrated near zero and only
 1068 a few reaching values up to 10. The concentration bound shows that even for the largest observed
 1069 biases, a sample size of $n = 10,000$ yields estimates of $\hat{\delta}_n(k)$ that deviate from the true value by
 1070 no more than a small ε with high probability. This justifies our sampling strategy and confirms that
 1071 accurate bias measurements are attainable with limited data.

1072 J Monotonicity and Calibration-Free Interpretation of δ

1073 Our goal when analyzing blind spots is to rank concepts by the severity of their generative bias. In
 1074 practice, we use the score $\delta(k)$ for this purpose. However, one may wonder whether such a score
 1075 introduces distortions relative to more direct quantities such as the energy gap or the odds ratio.
 1076 The following result establishes that $\delta(k)$ is a strictly increasing reparameterization of both, and
 1077 therefore inherits their ordering. This guarantees that no calibration is needed when using $\delta(k)$ to
 1078 rank concepts.

1079 **Theorem 5** (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy*
 1080 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_{\mathcal{X}}}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{\mathcal{X}}}[\xi_k(\mathbf{x})],$$

1081 *the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1082 *Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of*
 1083 *$\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.*

1084 *Proof.* The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore
 1085 $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and
 1086 bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting
 1087 this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two
 1088 strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict
 1089 monotonic functions never reverse inequalities, the three quantities share the same total order over
 1090 concepts.

1091

□

1092 Thus, ranking concepts by $\delta(\cdot)$ is strictly equivalent to ranking them by energy gap or by conceptual
 1093 generation odds ρ_k . No calibration is necessary, and all three quantities preserve the same total
 1094 ordering over concepts.

1095 K Stability of FID Under SAE Embeddings

1096 In this section we establish a quantitative relationship between the Fréchet Inception Distance
1097 (FID) computed in the original activation space (of dimension d) and the FID after applying a
1098 (potentially overcomplete) SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. Throughout we assume that D
1099 has orthonormal columns but is not necessarily square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

1100 We start by recalling a simple fact: if D is not overcomplete, orthogonal and $k = d$, then we have an
1101 isometry between A and Z , implying that the FID is perfectly preserved. However, this case is not
1102 realistic, we will then turn to the overcomplete case, and show we can bound FID by the extreme
1103 singular value of D . We will work with the Wasserstein-2 metric $\mathcal{W}_2$, noting that FID is just $\mathcal{W}_2^2$
1104 specialised to Gaussians.

1105 For a probability measure μ on $\mathbb{R}^d$ we write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) =$
1106 $\mu(D^{-1}z)$. Denote by $\sigma_{\min}$ and $\sigma_{\max}$ the minimal and maximal singular values of D , equivalently
1107 the square-roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

1108 Empirically one usually finds $\sigma_{\min}, \sigma_{\max} \approx 1$, but the proof does not rely on that. We will start by a
1109 simple lemma in the case where D is not overcomplete.

1110 **Lemma 1** (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then*
1111 *D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν*
1112 *on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

1113 *Proof.* Orthogonality of D implies preservation of the Euclidean norm, and push-forward commutes
1114 with the map inside the $\mathcal{W}_2$ infimum; the integrand is unchanged, so the infimum value is identical.
1115 □

1116 This case, however, is quite unrealistic as SAE usually rely on the overcompleteness to extract
1117 meaningful and interpretable concepts. In the overcomplete case, D is no longer orthonormal, but
1118 we can still have column-orthonormal dictionary. We will use that to show that we can bound using
1119 the extremal singular value of $D^\top D$.

1120 **Theorem 6** (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and*
1121 *denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices*
1122 *$A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

1123 *Proof.* Write μ for the empirical measure of A and ν for that of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

1124 For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have,
1125 by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

1126 Integrating with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

1127 The middle integral is exactly the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between
1128 $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they hold in particular

1129 for the optimal couplings attaining $\mathcal{W}_2(\boldsymbol{\mu}, \boldsymbol{\nu})$ and $\mathcal{W}_2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D)$, though these two optima need not
 1130 coincide. Taking the infimum over $\boldsymbol{\pi}$ term-wise makes this explicit:

$$\sigma_{\min}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi} \leq \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|D(\mathbf{x} - \mathbf{y})\|_2^2 d\boldsymbol{\pi} \leq \sigma_{\max}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi}.$$

1131 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}) \leq \mathcal{W}_2^2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

1132 Recognising $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and plugging in $(\mathbf{A}, \mathbf{A}')$ (resp. $(\mathbf{Z}, \mathbf{Z}')$)
 1133 finishes the proof. $\square$

1134 Essentially, theorem [6](#) tells us that applying a column-orthogonal overcomplete SAE dictionary
 1135 cannot distort Fréchet Inception Distance by more than the square of its extremal singular values.
 1136 When $\mathbf{D}$ is nearly orthogonal – empirically we usually found that $\sigma_{\min}, \sigma_{\max} \approx 1$ – the result implies
 1137 that FID measured in the SAE feature space is essentially close to the canonical FID. $\square$

1138